

# SIVABHARATHI M

## Machine Learning Engineer | AI & NLP Specialist

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### PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building NLP intent-classification models, computer vision and IoT-based detection systems, and production-style REST APIs. Skilled in the full ML pipeline — data preprocessing, model training, evaluation, and deployment — using Python, TensorFlow, and Scikit-learn. Strong software engineering foundation from full-stack development, enabling end-to-end delivery of AI-powered applications.

### TECHNICAL SKILLS

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|---------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Programming:</b> | Python, JavaScript, SQL, HTML5, CSS3                                                                      |
| <b>AI / ML:</b>     | Machine Learning, Deep Learning, NLP, Intent Classification, Model Training & Evaluation, Computer Vision |
| <b>Frameworks:</b>  | TensorFlow, Scikit-learn, NLTK, Flask, SQLAlchemy                                                         |
| <b>Databases:</b>   | PostgreSQL, SQLite                                                                                        |
| <b>Tools:</b>       | Git, GitHub, VS Code, Jupyter Notebook, REST APIs, Postman                                                |
| <b>Concepts:</b>    | OOP, Data Structures & Algorithms, API Integration, Model Deployment, Full-Stack Development              |

### AI / ML PROJECTS

#### Banking Chatbot — NLP Intent Classification System

*Python, NLTK, Scikit-learn, Flask, REST API*

- Built an NLP-based intent classification model trained to recognize 50+ banking-related query types using tokenization, stemming, and bag-of-words / TF-IDF feature extraction.
- Designed the text preprocessing and model evaluation pipeline, then deployed the trained model behind a Flask REST API integrated with a web front end.

#### Smart Security Surveillance System — IoT + Real-Time Detection

*ESP32, PIR & Sound Sensors, Telegram Bot API, Python*

- Engineered an IoT-enabled detection system combining motion and sound sensors with an ESP32 microcontroller for real-time event classification and alerting.
- Integrated Telegram Bot API for instant remote notifications, demonstrating applied signal-processing and event-driven inference on embedded hardware.

#### E-Commerce Platform — Full-Stack Application with Data Pipeline

*Flask, PostgreSQL, SQLAlchemy, JavaScript*

- Developed a full-stack platform with user authentication, product catalog, cart, order management, and an admin dashboard, backed by a relational PostgreSQL data layer.
- Built the REST API and database schema that would serve as the data backbone for future ML-driven features such as recommendation and demand forecasting.

### EDUCATION

#### B.Sc. Computer Science — Artificial Intelligence & Machine Learning

*NGM College, Pollachi | 2022 – 2025 | CGPA: 7.59 / 10*

### ACHIEVEMENTS

- Built multiple end-to-end AI, ML, NLP, and IoT projects spanning data collection, model training, and deployment.
- Delivered scalable applications integrating databases, REST APIs, and machine learning models.
- Strong grasp of the software development lifecycle, from problem framing to application deployment.